

By construction the output sets $\mathcal{Y}_1, \dots, \mathcal{Y}_d$ are pairwise disjoint and u_i is connected to $\mathcal{Y}_i$ by edges in $\mathcal{U}$. Also, for every $i \in \{1, \dots, d\}$ the algorithm adds at most $\frac{a}{2}u^{-\rho_-} + 1$ vertices to the graph $\mathcal{U}$, so that its output $\mathcal{U}$ satisfies

$$|\mathcal{U}| \leq |\mathcal{U}'| + d\left(\frac{a}{2}u^{-\rho_-} + 1\right) \leq a(m/u)^{\rho_-} \left(u^{\rho_-} + \frac{1}{2} + \frac{1}{a}u^{\rho_-}\right) \leq a(m/u)^{\rho_-},$$

for all $0 < u < u_0$ by choice of u_0 . In the following we show how the algorithm can be used to construct a suitably large subgraph of $\mathcal{G}_m$.

We run the algorithm with parameter $(\tilde{\pi}, u, m)$ for an intensity measure with a slightly decreased density parameter $0 < \tilde{\beta} < \beta$, $0 < u < u_0$ and some large m . This leads to a slightly smaller value of ρ_- which is referred to as ρ in the statement of Proposition 2. The next lemma shows that the probability of edges inserted by the algorithm is bounded from above by the edge probabilities in $\mathcal{G}_m$.

Lemma 2.3. *There exists $m(u) \in \mathbb{N}$ such that, for all $m \geq m(u)$, for all $m \geq s, r \geq bum$ with $s \neq r$ the probability that a particle v in location $V(v)$ with $\pi_m(V(v)) = r$ has an offspring y with location $V(y)$ satisfying $\pi_m(V(y)) = s$ is at most*

$$\beta(r \wedge s)^{-\gamma}(r \vee s)^{\gamma-1}.$$

Proof. For a fixed particle v in location $V(v)$ with $\pi_m(V(v)) = r$ the probability that it has an offspring y with location $V(y)$ satisfying $\pi_m(V(y)) = s$ equals

$$1 - \exp\left(-\tilde{\pi}\left(-\sum_{k=s}^m \frac{1}{k} - V(v), -\sum_{k=s+1}^m \frac{1}{k} - V(v)\right)\right). \quad (4)$$

As $\pi_m(V(v)) = r$ we have

$$-\sum_{k=r}^m \frac{1}{k} < V(v) \leq -\sum_{k=r+1}^m \frac{1}{k}.$$

The probability in (4) is therefore largest when $V(v) = -\sum_{k=r}^m \frac{1}{k}$. It therefore remains to show that, for $bum \leq s < r$, we have

$$1 - \exp\left(-\tilde{\pi}\left(-\sum_{k=s}^{r-1} \frac{1}{k}, -\sum_{k=s+1}^{r-1} \frac{1}{k}\right)\right) \leq \beta s^{-\gamma} r^{\gamma-1}, \quad (5)$$

and, for $bum \leq r < s$, we have

$$1 - \exp\left(-\tilde{\pi}\left(\sum_{k=r}^{s-1} \frac{1}{k}, \sum_{k=r}^s \frac{1}{k}\right)\right) \leq \beta s^{\gamma-1} r^{-\gamma}. \quad (6)$$

For (5) we find that, for some constant $C > 0$, if $m \geq m(u)$ for a suitable $m(u) \in \mathbb{N}$,

$$\begin{aligned} \tilde{\pi}\left(-\sum_{k=s}^{r-1} \frac{1}{k}, -\sum_{k=s+1}^{r-1} \frac{1}{k}\right) &= \frac{\tilde{\beta}}{1-\gamma} \exp\left(-(1-\gamma)\sum_{k=s}^{r-1} \frac{1}{k}\right) (e^{\frac{1-\gamma}{s+1}} - 1) \\ &\leq \left(\frac{\tilde{\beta}}{s+1} + \frac{C}{(bum)^2}\right) \exp\left(-(1-\gamma)\left(\log\left(\frac{r-1}{s-1}\right) - \frac{C}{bum}\right)\right) \\ &\leq \beta s^{-\gamma} r^{\gamma-1}. \end{aligned}$$

Hence, using that $1 - e^{-x} \leq x$, we get (5).